Now,

$$f'(x) = \frac{d}{dx}(\tan x) = \sec^2(x)$$

$$f''(x) = \frac{d}{dx}(\sec^2(x)) = 2\sec x \cdot \sec x \cdot \tan x = 2\sec^2 x \cdot \tan x$$

$$f'''(x) = \frac{d}{dx}(2\sec^2(x) \cdot \tan x) = 2\sec^2(x) \cdot \sec^2 x + \tan x \cdot 4\sec x \cdot \sec x \cdot \tan x$$

$$= 2\sec^4 x + 4\sec^2 x \cdot \tan^2 x$$

$$f^{(iv)}(x) = 8\sec^3 x \cdot \sec x \cdot \tan x + 4\sec^2 x \cdot 2\tan x \cdot \sec^2 x + 8\sec x \cdot \sec x \cdot \tan x \cdot \tan^2 x$$

$$= 16\sec^4 x \tan x + 8\sec^2 x \cdot \tan^3 x$$

$$f^{(v)}(x) = 16\sec^4 x \cdot \sec^2 x + 64\sec^3 x \cdot \sec x \cdot \tan x \cdot \tan x + 8\sec^2 x \cdot 3\tan^2 x \cdot \sec^2 x$$
$$+16\sec x \cdot \sec x \cdot \tan x \cdot \tan^3 x$$

$$= 16\sec^6 x + 88\sec^4 x \cdot \tan^2 x + 16\sec^2 x \cdot \tan^4 x .$$

Function and its derivatives	**$\tan x$ and its derivatives**	**value at $x = 0$**
$f(x)$	$\tan x$	0
$f'(x)$	$\sec^2 x$	1
$f''(x)$	$2\sec^2 x \tan x$	0
$f'''(x)$	$2\sec^4 x + 4\sec^2 x \cdot \tan^2 x$	2
$f^{(iv)}(x)$	$16\sec^4 x \cdot \tan x + 8\sec^2 x \cdot \tan^3 x$	0
$f^{(v)}(x)$	$16\sec^6 x + 88\sec^4 x \cdot \tan^2 x + 16\sec^2 x \cdot \tan^4 x$	16

Table 7.3

Substituting the values and on simplification we get the required expansion of the function as

$$\tan x = x + \frac{1}{3}x^3 + \frac{2}{15}x^5 + \cdots \quad ; -\frac{\pi}{2} < x < \frac{\pi}{2} .$$

Example 7.32

Write the Taylor series expansion of $\frac{1}{x}$ about $x = 2$ by finding the first three non-zero terms.

Solution

Let $f(x) = \frac{1}{x}$, then the Taylor series of $f(x)$ is

$$f(x) = \sum_{n=0}^{n=\infty} a_n (x-2)^n \text{ , where } a_n = \frac{f^{(n)}(2)}{n!} .$$

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